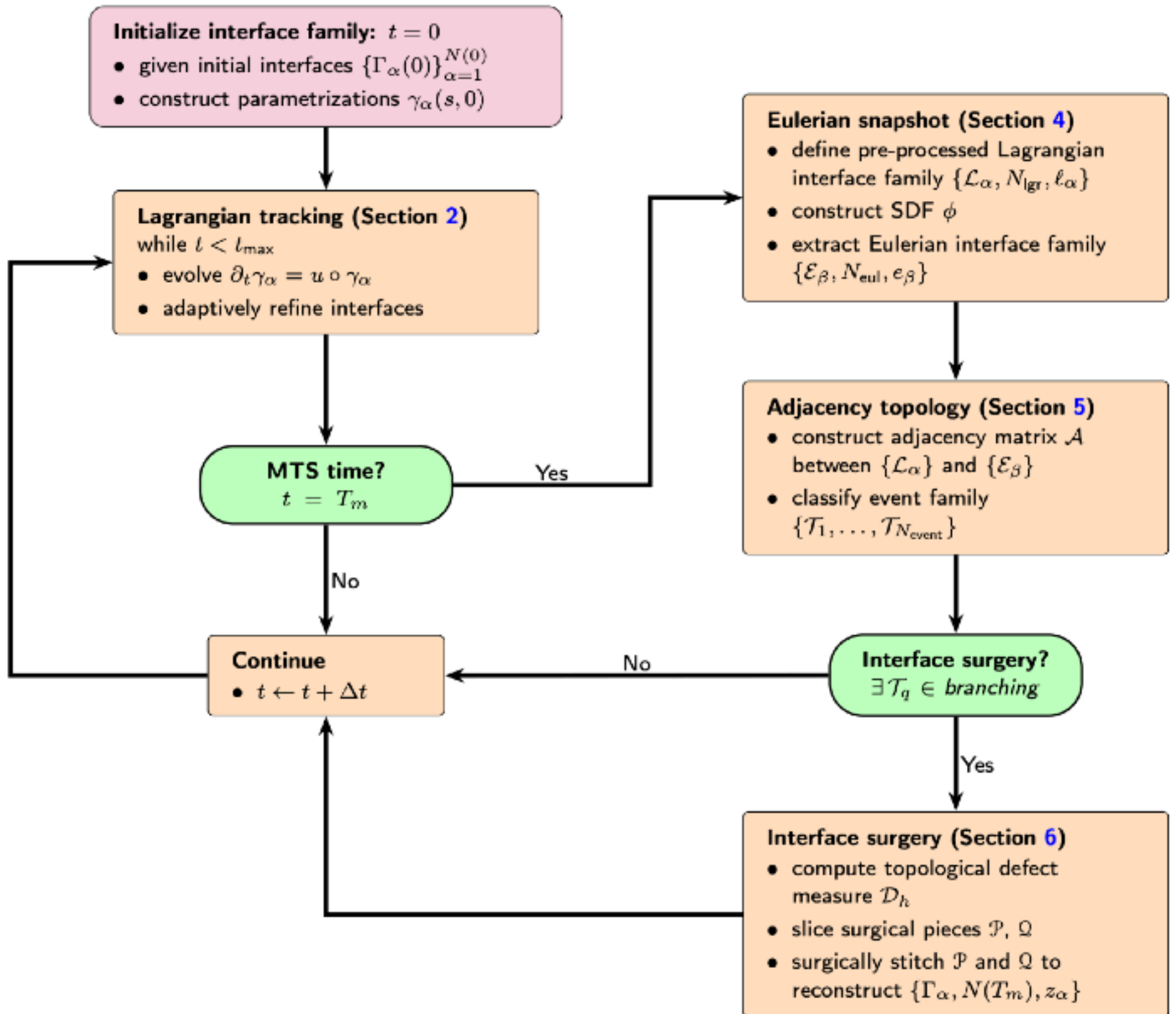


MICROSCALE TOPOLOGICAL SURGERY (MTS)



INTERMITTENT TOPOLOGICAL CHANGES VIA
GEOMETRIC REGULARIZATION AT MICROSCALE





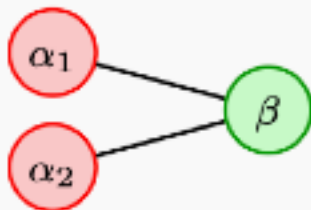
$$A_q = [1]$$

trivial



$$A_q = [0]$$

vanishing



$$A_q = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

merge



$$A_q = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

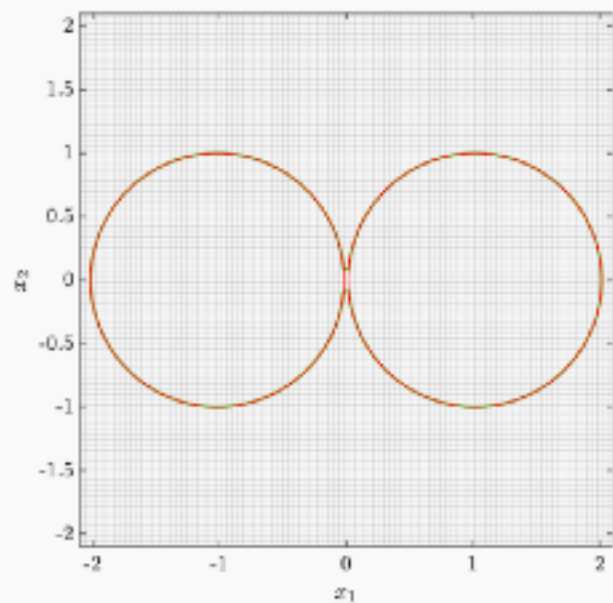
split



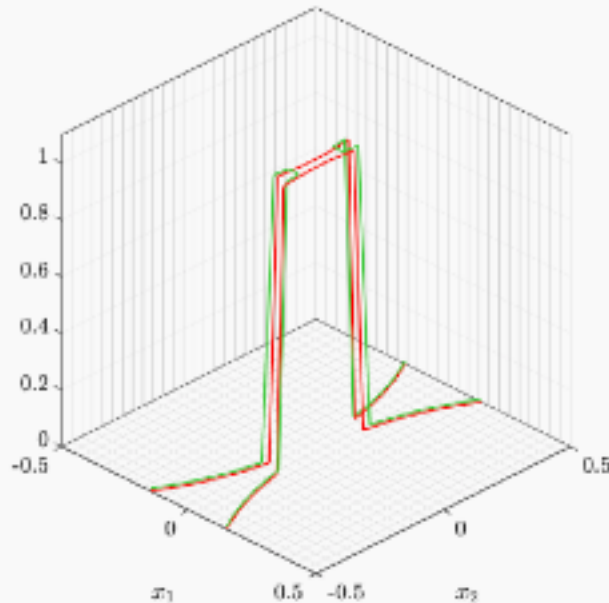
$$A_q = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

branching

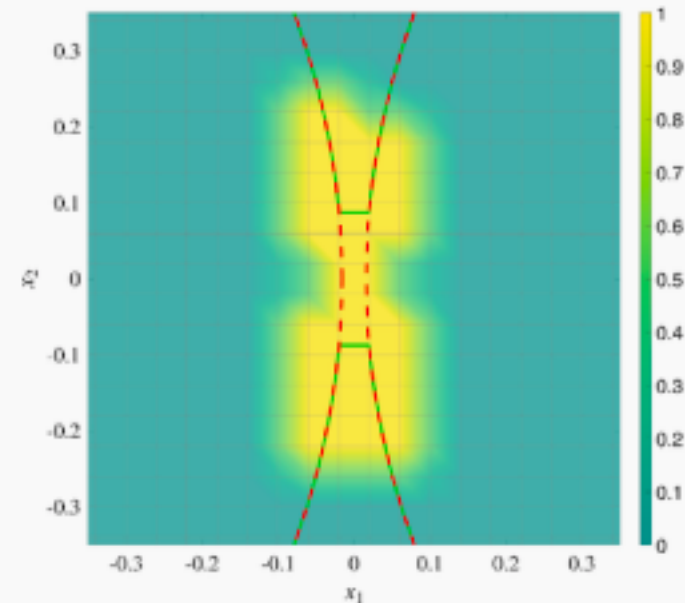
ADJACENCY TOPOLOGY. A bipartite graph encodes correspondences between Lagrangian & Eulerian interface families and classifies topological events.



(A) Interfaces $\mathcal{L}_\alpha$ and $\mathcal{E}_\beta$



(B) Defect densities (μ, η)

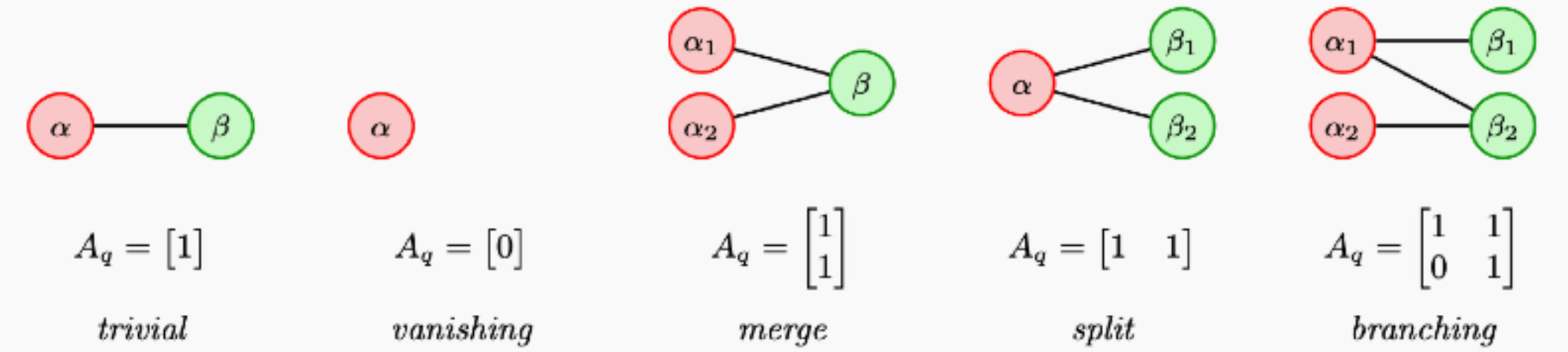
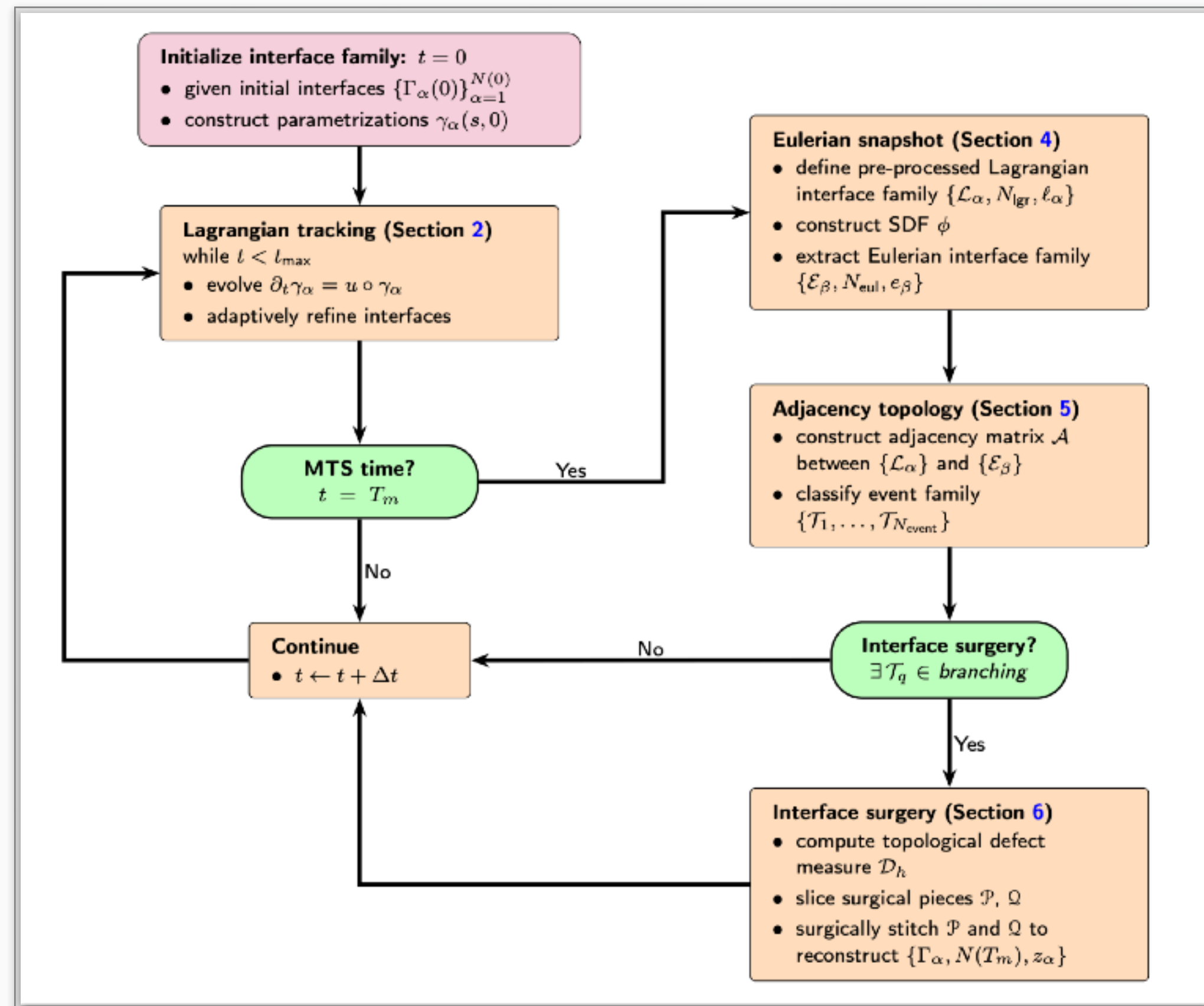


(C) Regularized defect measure $\mathcal{D}_h$

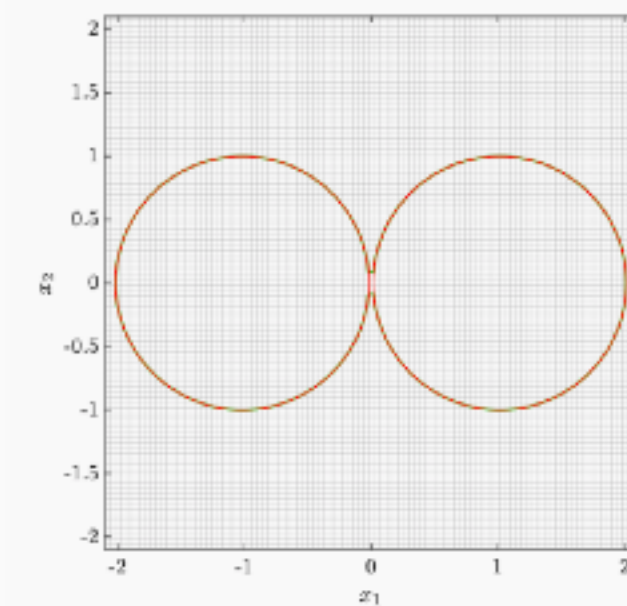
INTERFACE SURGERY. *Degeneracies in bidirectional correspondences are quantified by localized defect measures that identify surgical reconstruction regions.*

MICROSCALE TOPOLOGICAL SURGERY (MTS)

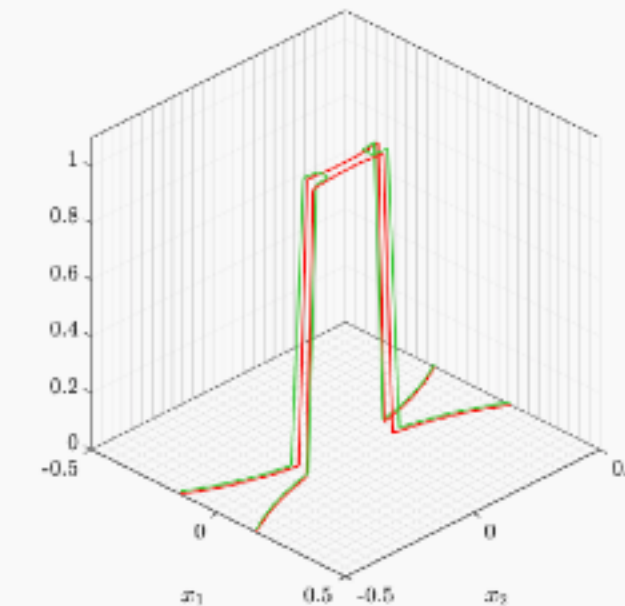
INTERMITTENT TOPOLOGICAL CHANGES VIA GEOMETRIC REGULARIZATION AT MICROSCALE



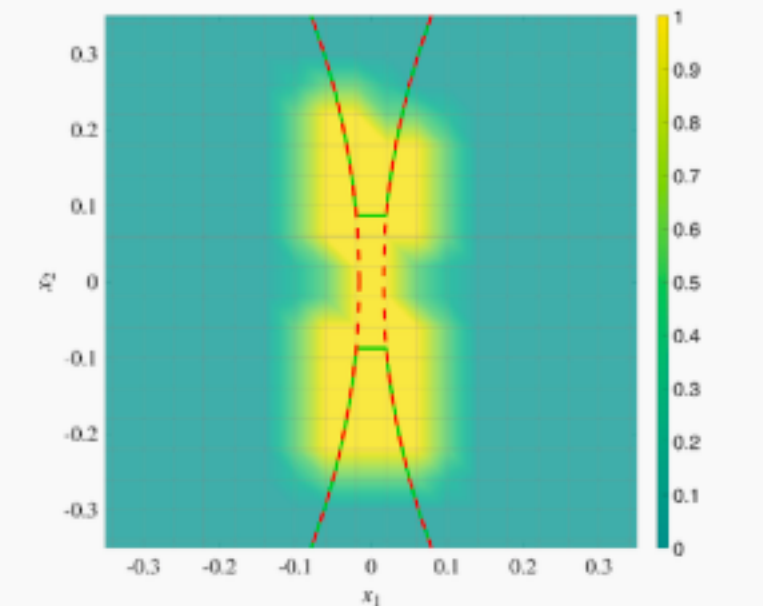
ADJACENCY TOPOLOGY. A bipartite graph encodes correspondences between Lagrangian & Eulerian interface families and classifies topological events.



(A) Interfaces $\mathcal{L}_\alpha$ and $\mathcal{E}_\beta$



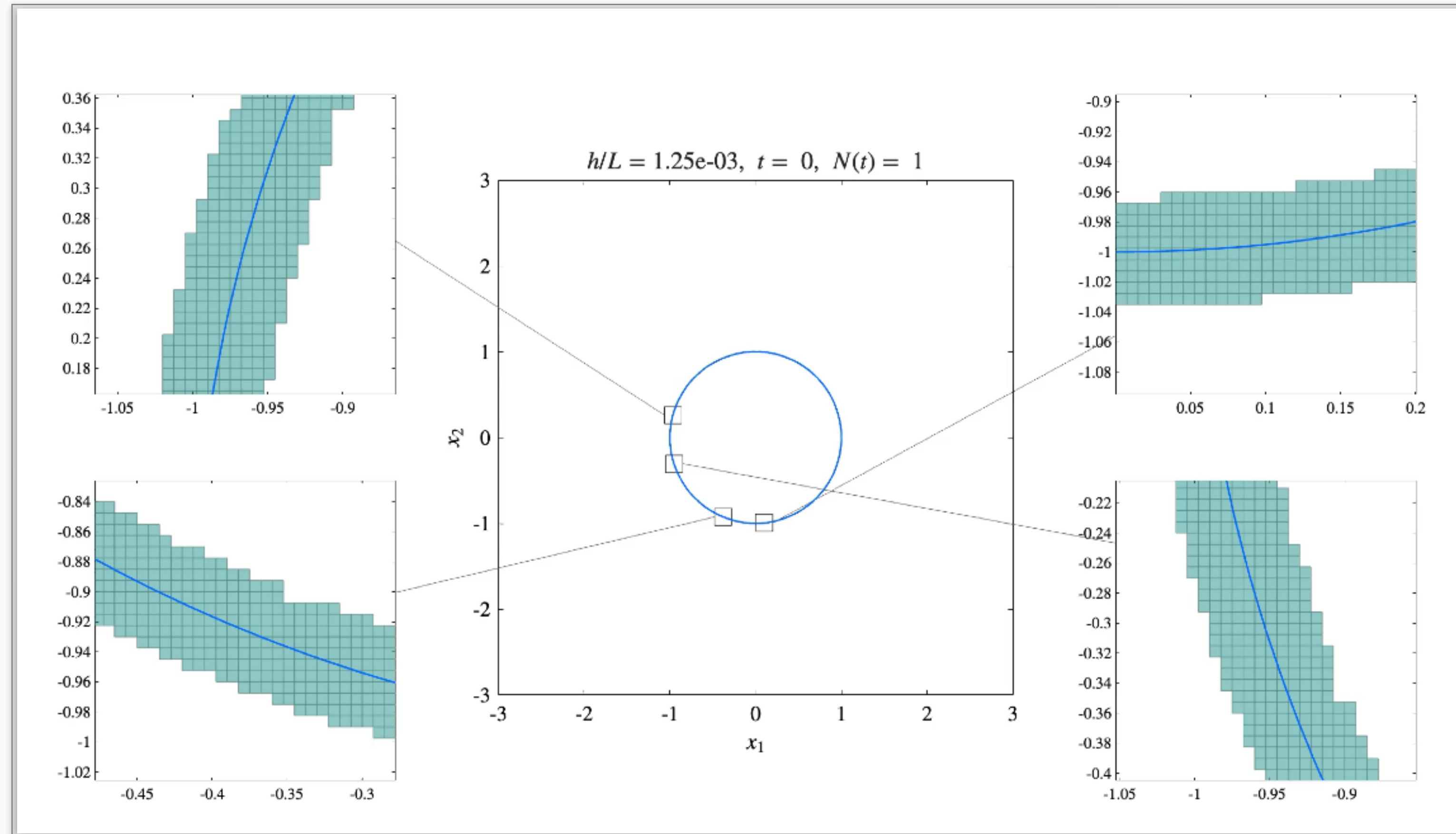
(B) Defect densities (μ, η)



(C) Regularized defect measure $\mathcal{D}_h$

INTERFACE SURGERY. Degeneracies in bidirectional correspondences are quantified by localized defect measures that identify surgical reconstruction regions.

MICROSCALE FILAMENT BREAKUP



SATELLITE INTERFACE DYNAMICS. As filament widths collapse, the MTS algorithm computes the geometry & topology of filament breakup through interface surgery, generating microscale satellites detached from the primary interface.